



Why the cloud needs observability

How observability can transform your IT operations



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INTRODUCTION

Why the cloud needs observability

Organizations today turn to cloud environments for their unparalleled benefits in terms of scalability, flexibility, and cost-effectiveness. However, they also bring a unique set of challenges that can be daunting to even the most seasoned IT professionals.

In modern cloud environments, observability is crucial for every element of cloud adoption. Whether migrating to the cloud, developing new applications, managing performance, ensuring security, controlling costs, or preparing for disaster recovery, observability provides the visibility and insights needed to make informed decisions and take proactive actions.

Observability is the practice of instrumenting systems to provide insights into their performance, behavior, and overall health. It's not just about monitoring metrics, traces, and logs. Teams also gain a comprehensive understanding of their entire cloud ecosystem. Observability enables cloud engineers and IT leaders to ensure their systems are functional and optimized for peak performance.

We will explore and offer optimization tips in the most common cloud scenarios where observability is crucial. Read on to discover how mastering observability unlocks the full potential of your cloud environment.

Observability foundations for successful cloud adoption

Migration. Observability identifies potential issues and bottlenecks during the migration process to ensure a smooth transition to the cloud.

Development. Real-time insights into the development pipeline ensure that code changes do not introduce new issues or degrade performance.

Performance. Continuously monitoring and analyzing performance metrics help to maintain optimal system performance and a positive user experience.

Security. Observability tools can detect and respond to security threats in real time, safeguarding sensitive data and maintaining compliance.

Cost management. Detailed insights into resource utilization help to manage costs.

Disaster recovery. In the event of a failure, observability provides the data needed to quickly identify the root cause and implement effective recovery strategies.



Weighing the importance of observability for cloud migration and adoption

Organizations are rapidly shifting their operations to the cloud for its scalability, flexibility, and cost-effectiveness. But this migration comes with its own set of challenges. How do you ensure a seamless transition without compromising performance or user experience? The answer lies in observability.

From lift-and-shift approaches to re-platforming and refactoring, businesses employ various strategies to make this shift. Each method has its merits and pitfalls, but understanding and navigating the complexities of monolithic applications remains a pivotal starting point.

Cloud migration challenges

Understanding your legacy environment

Migrating from a monolithic architecture to microservices can be daunting. The first step is to thoroughly understand the applications in question. Monolithic applications are complex and tightly coupled, making it difficult to isolate and migrate individual components without disrupting the entire system. Consider utilizing the following best practices to assist in migrating monolithic applications:

- **Dependency mapping.** Identifying all dependencies within a monolithic application is challenging but crucial. Automated topology mapping can simplify this process, providing a clearer picture of how different components interact.
- **Telemetry data.** Utilize request traces and telemetry data to understand the application's behavior and dependencies, facilitating a smoother transition to microservices.

Finding the right candidates for refactoring

Not all parts of a monolithic application are suitable for immediate refactoring. Identifying the right candidates is essential for minimizing disruption and maximizing efficiency. Use the following approaches to determine optimal refactoring candidates:

- **Domain-driven design.** Separate microservices based on their underlying business functions. This approach allows for more manageable and effective refactoring.
- **High business value components.** Start with components that have high business value and are loosely coupled. This strategy reduces the risk of performance issues and enhances user experience.

Identifying performance issues

Incorrectly executed migrations can lead to significant performance issues, negatively affecting user experience. Continuously monitoring and optimizing performance during and after migration is essential.

- **SLA, SLO, and SLI monitoring.** Use these metrics as performance benchmarks to ensure the migration doesn't degrade performance.
- **Automated dependency mapping.** Tools that automatically map dependencies can quickly identify performance degradation caused by new releases.

Observability solutions to cloud migration problems

Observability is a continuous process that's essential for every element of cloud adoption. By implementing observability before, during, and after a migration, teams can better understand applications and maintain performance standards.

A well-equipped observability solution should provide the following:



Automated topology mapping

Automatically map out the application's topology to understand dependencies and interactions. This step is crucial for planning the migration strategy.

What's more, real-time monitoring in addition to automated topology mapping can help keep an eye on the application's performance at all times to catch and resolve issues as they arise.



Cost-effective visualization

Visualizing an application's dependencies doesn't have to break the bank. There are cost-effective tools available that offer robust visualization capabilities, including the following:

- **Foundation and discovery.** These tools provide an inexpensive way to visualize dependencies and understand the bare bones of your environment.
- **Unified observability platforms.** These comprehensive observability solutions include logs, metrics, traces, and user experience monitoring, all in one place.



Full context understanding

End-to-end observability provides a complete picture of an application's performance and health. Service relationships help teams understand how different services interact with each other.

Application dependencies, meanwhile, offer insights into the dependencies within an application. This enables teams to make informed decisions and implement effective remediation strategies.

CUSTOMER SUCCESS STORY

“Migrating our core banking platform to the cloud was akin to performing open heart surgery whilst you’re doing a sprint. **The observability and telemetry Dynatrace provided was critical to making that project a success.**”

– Nic Olivier, Head of Platforms



CHAPTER TWO

Optimizing performance in diverse cloud ecosystems

Maintaining high cloud performance is paramount. However, as cloud architectures become increasingly complex, managing them effectively poses significant challenges. Microservices complexity, complicated private and public cloud management, evolving cloud service models, hybrid and multicloud challenges, and serverless computing difficulties are just a few examples of common hindrances.

Enter observability platforms, which help to ensure optimal cloud performance across diverse ecosystems. In this section, we'll explore how observability can optimize performance in various cloud environments, providing valuable insights for cloud engineers, IT leaders, and tech innovators.



CLOUD PERFORMANCE CHALLENGE

Multicloud complexity

Managing a multicloud environment is inherently complex. Each cloud provider has its own set of tools and monitoring systems, making it difficult to gain a holistic view of an entire infrastructure. Without end-to-end visibility, identifying performance bottlenecks becomes an uphill battle.



OBSERVABILITY SOLUTION

End-to-end visibility

Observability provides end-to-end visibility across different cloud environments in real time. A unified platform consolidates data from various sources, providing a comprehensive view of an entire cloud ecosystem. This holistic visibility is crucial for identifying performance issues and understanding their root causes.

CLOUD PERFORMANCE CHALLENGE

Dynamic and scalable architectures

Cloud environments are designed to be dynamic and scalable, allowing resources to be adjusted based on demand. While this is a significant advantage, it also becomes challenging to keep track of resource utilization and performance metrics in real time. Automation is essential to manage these dynamic environments effectively.



OBSERVABILITY SOLUTION

Proactive problem detection

Modern observability solutions leverage AI and machine learning to detect anomalies and potential issues proactively. Automated alerts and AI-driven insights help IT teams address problems before they affect performance. For example, Dynatrace uses its AI capabilities to enable real-time detection of performance degradation, allowing for swift remediation.

CLOUD PERFORMANCE CHALLENGE

Distributed architectures and bottlenecks

Distributed architectures, such as microservices, spread components across multiple servers and even different cloud environments. This distribution makes it difficult to pinpoint the source of performance issues. Traditional monitoring tools often fall short in providing the granularity needed to identify and resolve these bottlenecks.



OBSERVABILITY SOLUTION

Resource optimization

Observability tools provide insights into resource utilization, helping organizations optimize their cloud infrastructure. By understanding patterns and trends, teams can make informed decisions about resource allocation, ensuring the cloud environment is both cost-effective and performant.

CUSTOMER SUCCESS STORY

“Before Dynatrace, keeping up with our increasingly complex infrastructure was draining our teams’ time. **Now, we can automatically capture every source of observability and user-experience data in a single place.**”

— **Sean Mack**, Chief Information Officer and Chief Information Security Officer
WILEY

CHAPTER THREE

Expanding DevOps and platform engineering

DevOps and platform engineering make a strong case for cloud migration due to their ability to deliver better software faster. By integrating development and operations, these methodologies eliminate silos and improve cross-team collaboration. This leads to accelerated development cycles, increased reliability, and efficient resource utilization. Implementing a strong DevOps practice can reap several benefits for driving business value, including the following:

- **CI/CD pipelines.** Continuous integration and continuous delivery enable rapid iteration and deployment, ensuring that software is always in a releasable state.
- **Accelerated development cycles.** Faster feedback loops allow teams to identify and resolve issues quickly, improving overall productivity.
- **Increased reliability and scalability.** Cloud-native development ensures applications are built to scale efficiently, meeting the demands of modern users.
- **Efficient resource utilization.** By leveraging cloud resources, organizations can optimize costs and improve performance.

Cloud-native development allows teams to build and deploy applications that can fully leverage the benefits of the cloud. By incorporating telemetry through frameworks like OpenTelemetry, teams can gain deeper insights into application performance, user behavior, and system health.



DevOps and platform engineering challenges

Despite the numerous benefits, DevOps and platform engineering come with their own set of challenges:



Complexity

Managing multiple microservices, environments, and tools can be overwhelming. Teams may not have the bandwidth to adequately monitor and maintain their technology stack through manual effort alone



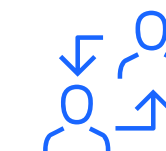
Security

Ensuring that security measures are integrated at every stage of the development lifecycle is critical but challenging. The often massive and complex nature of cloud-environments and the rising prevalence of Kubernetes infrastructures can make it extremely difficult to keep track of all the potential vulnerabilities that might arise, and which ones should be prioritized for mitigation.



Siloed automation

Automation efforts often become siloed within specific teams, leading to inefficiencies. When teams work from separate datasets or utilize different tools, automation workflows become increasingly convoluted and inefficient.



Cultural alignment

Aligning the diverse teams involved in DevOps methodology requires significant cultural shifts and collaboration. Many teams might resist any significant changes in organizational approaches and workflows, instead desiring the status quo of operations.

Observability solutions to DevOps and platform engineering problems

Observability offers a comprehensive solution to these challenges by providing a unified platform that enhances visibility, security, and collaboration.

A unified observability platform integrates monitoring, logging, and tracing capabilities, providing a single source of truth for all teams. This holistic view allows organizations to achieve the following:

- Gain real-time insights into system performance and user behavior.
- Integrate security measures at every stage of the development lifecycle, from code commit to production deployment.
- Break down silos and improve communication across development, operations, and security teams.

By opting for an end-to-end, unified observability platform, organizations centralize views and datasets across teams while ensuring that no aspect of their environment goes unmonitored. This enables the acceleration of issue and vulnerability detection, UX improvements, and cross-functional alignment.



CUSTOMER SUCCESS STORY

Let's look at a real-world example of how observability has transformed a business.

“By providing advanced observability across our DevOps processes and app delivery pipelines, Dynatrace allows us to see the smoke before the fire starts to burn. We can stop bad code from ever reaching production.”, says Simon Pilar, Director of DevOps & Toolchain at Clario

Clario's success story highlights the tangible benefits of implementing observability solutions. By gaining real-time insights into their systems, Clario was able to proactively manage potential issues, ensuring their software remained reliable and high performing.

Combating cloud security challenges with observability

Security breaches are on the rise, with bad actors continually finding new ways to exploit vulnerabilities. The cloud, while offering unparalleled scalability and flexibility, demands an entirely different security strategy. Unlike on-premises systems, cloud environments are dynamic, distributed, and highly complex. This complexity makes it easier for misconfigurations and vulnerabilities to go unnoticed, providing ample opportunities for malicious actors.

Adding to the challenge, there's a significant talent gap in the cybersecurity field. Many professionals lack the expertise to fully comprehend the intricacies of cloud security, leaving organizations exposed to various threats. These threats are not only increasing in number but are also becoming more sophisticated, requiring advanced solutions that go beyond traditional security practices.

Cloud security challenges



Misconfigurations

Misconfigurations are among the leading causes of cloud security breaches. These errors are often unintentional, stemming from the inherent complexity of cloud environments. Simple mistakes, such as improperly setting access controls or failing to secure storage buckets, can have catastrophic consequences.



Zero-day attacks

Zero-day attacks also pose a significant risk. These attacks are particularly challenging, as they target flaws that have yet to be patched or even discovered by security teams. In a cloud context, where applications and services are continually updated and deployed, the window of exposure can be alarmingly wide.



Dynamic environments

Additionally, the dynamic nature of cloud environments means threats are constantly evolving. Attackers employ advanced tactics such as lateral movement, where they exploit one compromised service to gain access to others within the same environment. This fluidity makes it difficult for static security measures to keep pace, necessitating more adaptive and responsive solutions.

Observability solutions to cloud security problems



Runtime and real-time application protection

Observability provides a comprehensive view of the entire cloud environment, enabling real-time monitoring and protection. By leveraging runtime application self-protection (RASP) and other real-time security measures, organizations can detect and mitigate threats as they occur. This proactive approach significantly reduces the risk of successful attacks and minimizes potential damage.



Security analytics

Advanced security analytics play a pivotal role in identifying and responding to threats. Observability platforms integrate with security information and event management (SIEM) systems to provide deep insights into user behavior, network traffic, and application performance. These insights enable security teams to detect anomalies and potential threats before they can escalate.



Cloud-native application protection platforms (CNAPP)

Cloud-native application protection platform (CNAPP) solutions are designed to address critical security issues in cloud environments, including misconfigurations, vulnerabilities, and any system compromises. By continuously monitoring cloud infrastructure for compliance with security best practices and regulatory requirements, CNAPP tools help organizations identify and remediate these issues. This automated approach not only enhances security but also ensures that configurations remain consistent and secure over time. Additionally, CNAPP solutions integrate with CI/CD pipelines, enabling continuous security throughout the development lifecycle.

Gartner highlights the importance of CNAPPs, stating that “by 2029, **60% of enterprises** that do not deploy a unified CNAPP solution within their cloud architecture will lack extensive visibility into the cloud attack surface and consequently fail to achieve their desired zero-trust goals.”

Implementing a CNAPP is arguably the most crucial step in establishing a robust cloud security posture.

CUSTOMER SUCCESS STORY

“Dynatrace gives us an uncompromising level of control over the security of our software by continuously monitoring for application vulnerabilities in our production environments. That means we instantly know how a new vulnerability affects our digital services, so we can respond quickly to keep our customers’ data safe.”

— **Luca Domenella**, Head of Cloud Operations and DevOps



CHAPTER FIVE

Ensuring cloud resiliency for business continuity

Cloud resiliency is synonymous with business continuity. Organizations that rely heavily on cloud infrastructure must have robust plans in place to withstand unexpected events. This year alone, notable cloud outages from AWS and Microsoft Azure have highlighted vulnerabilities even among the giants of cloud service providers. Security incidents have also underscored the need for thorough preparedness.

While architectural best practices like redundancy, scalability, and demand planning are fundamental, observability can significantly enhance their effectiveness. Observability provides deep insights into these practices, ensuring they work as intended. Disaster recovery plans are also essential, and observability must be a part of that plan to ensure holistic visibility of all compromised components.



Cloud resiliency challenges

Cloud environments are inherently complex. Multiple layers of services, microservices, and containers create a labyrinthine architecture that's difficult to oversee. Additionally, identifying which issues to tackle first in a sea of alerts can be daunting. Without clear visibility, prioritizing becomes guesswork.

In the event of an outage or security incident, speed is of the essence. Rapid detection and resolution are paramount to minimizing impact.

Observability solutions to cloud resiliency problems

Proactive planning and testing

Having a plan is essential, but testing its limits before an incident occurs is crucial. Observability helps organizations understand how a system reacts under stress and identifies potential failure points. For example, Dynatrace enabled a U.S.-based pharmaceutical company to recover critical systems within hours of the CrowdStrike incident, thanks to real-time monitoring and the Problems application that identified issues and their root causes swiftly.

Automation for speed

Automation is a critical component of observability. Automated analysis and alerting, often powered by AI, enable real-time visibility into cloud applications and user impact. Comprehensive application performance and root-cause analysis tools help troubleshoot incidents before they escalate. For instance, during the CrowdStrike crisis, synthetic monitoring provided early warnings, allowing companies to act swiftly and mitigate disruption.

Prioritization and remediation

Observability solutions make it easier to understand which services are most critical, allowing organizations to prioritize and remediate effectively. Dynatrace Smartscape topology mapping, for instance, offers dynamic, real-time visualization that helps teams prioritize remediation efforts based on the mission-critical nature of affected services.



CUSTOMER SUCCESS STORY

A North American telecommunications provider successfully navigated the CrowdStrike update crisis using the Dynatrace Query Language (DQL) and Dynatrace Notebooks. These tools enabled the provider to create custom charts and identify which systems were no longer forwarding logs. This newfound visibility allowed the team to prioritize recovery efforts, restoring essential services promptly.

CONCLUSION

Master the cloud with observability

In the intricate world of cloud computing, observability is not just a luxury—it's a necessity for ensuring resiliency. From proactive planning and automation to intelligent prioritization, observability empowers organizations to detect, analyze, and resolve issues swiftly and efficiently.

By integrating observability into your cloud strategy, you don't just prepare for the worst—you ensure your business can thrive even in the face of unexpected challenges. Remember, the key to cloud resiliency lies in your ability to see clearly through the complexities of your environment and act decisively.

Dynatrace's unified observability platform provides a comprehensive view across your entire technology stack with built-in security analytics and protection, acting as a one-stop shop for all your observability needs in the cloud. As a single source of truth with easy deployment via a single agent, Dynatrace reduces cloud complexity by providing end-to-end context and answers from metrics, logs, and traces.

Mastering the cloud with observability is a strategic imperative for modern enterprises. By leveraging Dynatrace's end-to-end, unified observability across these five critical scenarios—cloud adoption and migration, performance in diverse ecosystems, DevOps and platform engineering, security, and resiliency—organizations can unlock unparalleled efficiency and innovation.

Automatic and intelligent observability for hybrid multiclouds

We hope this ebook has inspired you to take the next step in your digital journey. Dynatrace is committed to providing enterprises the data and intelligence they need to be successful with their enterprise cloud and digital transformation initiatives, no matter how complex.

Learn more

If you are ready to learn more, please visit www.dynatrace.com/platform for assets, resources, and a **free 15-day trial**.



[Dynatrace](#) (NYSE: DT) exists to make the world's software work perfectly. Our end-to-end platform combines broad and deep observability and continuous runtime application security with [Davis® hypermodal AI](#) to provide answers and intelligent automation from data at an enormous scale. This enables innovators to modernize and automate cloud operations, deliver software faster and more securely, and ensure flawless digital experiences. That's why the world's largest organizations trust the Dynatrace® platform to accelerate digital transformation.

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